

Marwan Abdelaal

Backend and Full-Stack Developer | AI Engineering Intern | Software Engineering Student

marwan.abdelaal@outlook.com | +201111210005 | Cairo, Egypt | linkedin.com/in/marwan-abdelaal | github.com/MoscowAbdelaal

SUMMARY

3rd-year Software Engineering student at GIU (GPA: 2.4, Very Good, German Scale) with a Minor in IT Security. Currently interning as a Backend AI Engineering Intern at FlyRank AI, where I have shipped production-ready REST APIs, persistent database systems, JWT authentication, web scraping pipelines with local AI enrichment, automated test suites with 100% pass rates, and a full capstone multi-platform social campaign publisher with idempotency, durable scheduling, and HMAC-verified webhooks. Proficient in Node.js, Express.js, Python, SQL, PostgreSQL, Docker, Redis, and React.js, with hands-on experience in AI integration, cryptography, distributed systems, machine learning, and prompt engineering. Led a team of 10 to ship an AI-powered MERN platform. Holder of 20 Anthropic Academy certifications.

SKILLS

Backend:	Node.js Express.js REST APIs JWT Middleware Supabase Auth Swagger UI BullMQ
Languages:	JavaScript Python Java C# SQL HTML and CSS
Databases:	SQLite PostgreSQL MongoDB Microsoft SQL Server Redis Schema Design Query Optimization
AI and ML:	Hugging Face Ollama NLP Prompt Engineering Embeddings Naive Bayes KNN K-Means PCA
Security:	AES-GCM HMAC 3DES RSA Diffie-Hellman OAuth JWT SQL Injection Prevention
Reliability:	Idempotency Rate Limit Handling Durable Scheduling Webhook Verification Retry Logic Backoff
Data and Scraping:	Cheerio Axios Zod Sharp pandas scikit-learn matplotlib Jupyter Notebook
DevOps:	Docker Docker Compose Git GitHub GitLab CI/CD
Frameworks:	React.js JavaFX Unity .NET
Distributed:	MapReduce Weak Consistency Distributed Locking Synchronization Adapter Pattern
Testing:	Automated Testing Postman Manual Test Cases Parameterized Queries API Validation
Soft Skills:	Leadership Communication Problem-Solving Agile Scrum Time Management

PROFESSIONAL EXPERIENCE

Backend AI Engineering Intern - FlyRank AI	Jun 2026 - Present Remote
* Built a production-ready Node.js/Express REST API with 12 endpoints, full CRUD, SQLite persistent storage, parameterized queries for SQL injection prevention, and 23 automated tests all passing with 100% success rate. * Implemented a secure JWT authentication system using Supabase Auth with reusable middleware, Swagger UI documentation, and proper 401/403 HTTP status distinction for protected route management. * Developed a production-grade web scraping pipeline collecting 60 books using Cheerio and Axios with Zod schema validation, exponential backoff retry logic, caching reducing re-run time from 45s to 1s, CSV export, and local AI enrichment via Ollama. * Migrated backend storage to PostgreSQL in Docker using a repository pattern that swaps storage without changing service or route logic, orchestrated via Docker Compose with proven data persistence across restarts. * Engineered a 5-iteration prompt ladder applying role assignment, context framing, few-shot examples, and output structure techniques, comparing outputs across Claude and ChatGPT to produce reusable prompt templates. * Completed 20 Anthropic Academy certifications covering AI fluency, Claude tooling, MCP, agent skills, subagents, Claude API, Amazon Bedrock, and Google Cloud integrations.	
Part-Time Teaching Assistant - ICTeam El Komy	Jun 2024 - May 2026 Cairo, Egypt
* Delivered technical guidance to 20+ students across lab sessions, simplifying complex concepts and troubleshooting issues in real time, strengthening communication and leadership skills. * Adopted an agile approach to adapt session content based on student feedback, ensuring continuous improvement across multiple class cycles.	
Event Volunteer and Usher - Various Organizations	2024 - Present Cairo, Egypt
* Coordinated operations across 3+ large-scale events including the Cairo International Film Festival and university career fairs, managing volunteer teams under high-pressure conditions.	

PROJECTS

Multi-Platform Social Campaign Publisher - FlyRank Capstone

Tech: Node.js, Express.js, BullMQ, Redis, PostgreSQL, Docker, Sharp, AES-GCM, HMAC | github.com/MoscowAbdelaal/flyrank-capstone-social-studio

- * Built a production-grade social publishing system converting a single blog post into a complete multi-platform campaign, generating platform-correct image variants (Instagram 1080x1080, X 1600x900) and platform-tailored captions from reusable prompt fragments.
- * Implemented a clean adapter architecture with a SocialPublisher interface decoupling the app from platform specifics, so adding a platform means adding one adapter without touching business logic.
- * Engineered idempotent publishing ensuring duplicate requests or post-timeout retries always yield exactly one post per platform, with 429/Retry-After backoff and exponential retry logic for rate limit handling.
- * Built a durable scheduler using BullMQ and Redis that survives crashes and worker restarts, with scheduled posts resuming without creating duplicates and OAuth tokens stored encrypted with AES-GCM and random IV.
- * Secured delivery webhooks with HMAC signature verification rejecting forged events with 400, with campaign status transitioning from queued to published only on verified webhook delivery.

GIU Nexus - AI-Powered Career and Talent Platform

Tech: React Vite, Node.js, Express.js, MongoDB, Hugging Face API, JWT, Nodemailer, GitLab | github.com/MoscowAbdelaal/GIU-Nexus

- * Led a team of 10 to build a full-stack MERN application with Hugging Face NLP for skill extraction and zero-shot classification, cosine similarity recommendations, JWT auth, and webhook-based notifications.
- * Optimized backend performance by improving query efficiency, reducing API response times, and validated all endpoints via Postman across functional, integration, and regression test scenarios.

CRUD API - Production REST API with Auth and Testing

Tech: Node.js, Express.js, SQLite, PostgreSQL, Docker, Supabase Auth, JWT, Swagger UI | github.com/MoscowAbdelaal/crud-api

- * Built a full REST API with 12 endpoints, persistent SQLite storage, 23 automated tests at 100% pass rate, JWT auth via Supabase, reusable middleware, and Swagger UI docs, then migrated to PostgreSQL in Docker using a clean repository pattern.

Clash of Clans: Heroes - 2-Player Strategy Game

Tech: Java, JavaFX, Eclipse IDE, OOP, CSV Data Loading | github.com/MoscowAbdelaal/Clash-of-clans-Legends

- * Independently built a complete 2-player turn-based strategy game across 3 milestones with full OOP architecture including abstract classes, interfaces, inheritance, custom exception handling, and a JavaFX frontend with real-time updates.

Human Resource Management System (HRMS)

Tech: Microsoft SQL Server, HTML, CSS, Node.js, SQL Functions and Stored Procedures

- * Led full EERD and relational schema design, implemented optimized SQL stored procedures as the backend layer, and delivered a multi-role enterprise system covering payroll, attendance, leave, hiring, and performance reviews.

Web Scraping Pipeline with AI Enrichment

Tech: Node.js, Cheerio, Axios, Zod, Ollama, HTTP Server

- * Production-grade scraper collecting 60 books with Zod validation, exponential backoff, polite rate limiting, caching from 45s to 1s, CSV export, change detection, and local AI enrichment via Ollama.

Cryptography Suite - 3DES and Diffie-Hellman/RSA

Tech: Python, Jupyter Notebook

- * Implemented Triple DES encryption from scratch using EDE structure across 16-round blocks, and built a secure communication system combining Diffie-Hellman key exchange with RSA digital signatures simulating HTTPS-like secure transmission.

Machine Learning - Classification and Clustering

Tech: Python, scikit-learn, pandas, matplotlib, Jupyter Notebook, Kaggle

- * Built Naive Bayes and KNN classifiers to predict metabolic syndrome risk with hyperparameter tuning and Kaggle leaderboard submission. Applied K-Means, GMM, BIRCH, and Agglomerative Clustering to 561-feature HAR sensor data with PCA visualization.

Distributed Systems - MapReduce and Theater Booking Simulation

Tech: Python, Google Colab

- * Implemented MapReduce for inverted index, temperature aggregation, movie ratings, and product sales. Built a distributed theater booking simulation with 4 parallel counters, weak consistency, local queuing, and synchronized global state using distributed locking.

Interactive Media Experience - Unity

Tech: Unity, C#, AudioSource, Animator, TextMeshPro

* Designed and built an interactive Unity experience with 4+ media types including graphics, animation, audio, and text, featuring 3+ meaningful user interactions and state-based media responses.

CERTIFICATIONS - ANTHROPIC ACADEMY (2026)

- | | |
|---|--|
| * AI Fluency: Framework and Foundations | * Claude 101 |
| * Claude Code 101 | * Claude Code in Action |
| * Claude Platform 101 | * AI Capabilities and Limitations |
| * AI Fluency for Students | * AI Fluency for Educators |
| * AI Fluency for Builders | * AI Fluency for Small Businesses |
| * AI Fluency for Nonprofits | * Teaching AI Fluency |
| * Introduction to Claude Cowork | * Introduction to Agent Skills |
| * Introduction to Subagents | * Introduction to Model Context Protocol |
| * Model Context Protocol: Advanced Topics | * Building with the Claude API |
| * Claude with Amazon Bedrock | * Claude on Google Cloud |

EDUCATION

Bachelor of Software Engineering - Informatics and Computer Science -
German International University (GIU)

GPA: 2.4 / Very Good (German Scale) | Minor: IT Security | Expected Graduation: 2028 | Coursework: Data Structures and Algorithms, OOP, Database Systems, Data Science, Information Security, Distributed Systems, Media Informatics

Sep 2024 - Present | Cairo, Egypt

LANGUAGES

Arabic (Native) | English (Fluent) | German (Beginner) | French (Beginner)

HOBBIES

Traveling | Swimming | Music | Fitness and Gym | Motorcycle Riding | Diving